Chemistry - Graduate Level Assessment 11 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the approximate boiling point at standard pressure of a solution prepared by dissolving 234 g ofNaClin 500 g of H_2 O?

   (a) 102.5

   (a) 100

   (a) 109.5

   (a) 106.78

   (a) 101.22

2. A substance that is NOT generally considered to be a toxic pollutant in water is

   (a) nitrogen
   (b) a halogenated hydrocarbon
   (c) chlorine
   (d) copper
   (e) arsenic

3. Of the following compounds, which has the lowest melting point?

   (a) NaCl
   (b) LiCl
   (c) AlCl 3
   (d) KCl
   (e) FeCl 3

4. 10.0 grams of urea, CO(NH_2)_2, is dissolved in 200 grams of water. What are the boiling and freezing points of the resulting solution?

   (a) Boiling point is 101.00$^{,}Freezingpointis1.55$

   (a) Boiling point is 101.00$^{,}Freezingpointis-2.00$

   (a) Boiling point is 99.57$^{,}Freezingpointis1.55$

   (a) Boiling point is 100.0$^{,}Freezingpointis0.0$

   (a) Boiling point is 100.43$^{,}Freezingpointis1.55$

5. Calculate the weight in grams of sulfuric acid in 2.00 liters of 0.100 molarsolution. (MW of H_2 SO_4 = 98.1.)

   (a) 10.0 g
   (b) 24.4 g
   (c) 15.2 g
   (d) 22.3 g
   (e) 19.6 g

True / False (10 marks)

Answer True or False

6. True or False: AlBr 3 requires a Roman numeral in its name due to the variable oxidation state of the cation aluminum.

7. True or False: Diluting 50 ml of 3.50 M H_2 SO_4 to achieve a concentration of 2.00 M requires a final volume of 125 ml.

8. True or False: The quotient [{2 x^0} / {(2 $x)^0$}] simplifies to x^2, which incorrectly applies the properties of exponents.

9. True or False: To produce 1.50 g of O_2, 8.45 g of NaNO_3 must be decomposed according to the balanced reaction.

10. True or False: The zero-point entropy of nitric oxide (NO) is estimated to be 0.00 cal mol$^{-}1K^{-}1, indicating no disorder at absolute zero.$

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the cell potential (E) for the electrochemical cell Ag/AgCl/Cl^{-}(0.0001)andCe^{4}+(0.01), Ce^{3}+(0.1)/Pt using the provided standard half-cell potentials. Discuss your approach and calculations.

12. Discuss the decomposition of solid mercury(II) oxide (HgO) into oxygen gas (O_2) and liquid mercury (Hg). Calculate the moles of O_2 produced from 4.32 g of HgO, considering its molar mass.

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